

Arpan Pathak

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📄 SUMMARY

Senior Systems Software Engineer, **8 years**: concurrency, distributed systems, parallel computing, high-performance computing (HPC), and performance engineering, spanning kernel (eBPF/Cilium) to GPU (CUDA) to production ML inference (XGBoost, BERT) at **Amazon, Microsoft, Oracle**. Full-stack **Rust, C++, Kotlin, Java, Go, Python, JavaScript**; shipped zero-trust networking, lock-free DB internals, real-time ML ranking.

🔗 PERSONAL PROJECTS

🚗 **Driving-CivicSense: Edge AI Vision System (Rust)** | github.com/arpanpathak/driving-civicsense-vision-model

- On-device **edge** perception in **Rust** on the **NVIDIA Jetson Orin Nano Super** (67 INT8 TOPS): **YOLOv8/v11 INT8 ONNX (~12 ms)** + Deep SORT/Kalman + proven kinematic engine; **privacy-first, 100%** on-device.
- Research paper: **Deterministic Intersection Blockage Prediction** | arpanpathak.github.io/driving-civicsense-vision-model/
- Books: **Seeing Machines** | arpanpathak.github.io/seeing-machines-book · **CUDA Kernels** (CUDA-Oxide / Rust-to-CUDA) | arpanpathak.github.io/gpu-parallel-book

🏢 EXPERIENCE

Software Developer II, Microsoft, Redmond, WA

Jun 2025 – Aug 2026

- **Zero-Trust Sovereign Cloud Networking**: designed and delivered zero-trust sovereign cloud networking: eBPF + Cilium policies enforcing **data residency** & **SecNumCloud** compliance; Kubebuilder operators across **AKS** clusters (France, Germany).
 - **C++/C# → Rust Migration (Microsoft Security)**: owned the migration of **Microsoft Information Protection** from unsafe C++/C# to Rust; mentored engineers through the transition; killed use-after-free/buffer-overflow/null bugs, **~40% lower P99, 2x throughput, 3x lower memory**, 6-figure savings.
 - **AI-Powered Policy Engine**: built an AI engine auto-generating Cilium policies and scanning misconfigurations; review **4 hours → near-real-time**, zero drift.
- 🔗 TechStack: **Rust, C++, eBPF, Cilium, Kubernetes (AKS), Kubebuilder, Prometheus, Linux**

Senior Software Engineer, Oracle Cloud Infrastructure, Seattle, WA

Oct 2024 – Mar 2025

- **High-Performance Database Engine**: designed and implemented **lock-free data structures** cutting contention in high-concurrency read-write paths; **45,000 ops/s** throughput gain.
 - **Infrastructure-as-Code SDK**: owned the **Terraform provider** (developer SDK) for Oracle Autonomous Database, replacing legacy control-plane tooling; **\$1.2M/month** projected savings.
 - **Security & Key Management**: integrated OCI Security Vault with automated key rotation across control plane services.
- 🔗 TechStack: **Java, Go, Terraform, OCI, Autonomous Database**

Software Development Engineer II, Amazon, Seattle, WA & Hyderabad, India

Mar 2021 – Oct 2024

- **Real-Time ML Ranking Engine**: owned end-to-end (training, serving, latency tuning): XGBoost + Deep Learning ensemble across **FreeVee, Twitch, MiniTV, Prime Video, Amazon Retail**; **5% CTR lift, \$5.4M/month revenue**.
 - **BERT Inference Service**: deployed and scaled **BERT-based ML inference** at **5,000 QPS** (ONNX Runtime, INT8 quantization, latency tuning); cut triage SLA **7 days → 2 hours**.
 - **Ads Publisher Platform (Event-Driven)**: led a team of 3 developers building event-driven systems for **programmatically guaranteed ad deals** via real-time **CDC** at **< 50ms** (Kotlin, CSP).
 - **NoSQL SDK**: designed a uniqueness-constraint indexing SDK (two-phase commit, **serializable isolation, 3K writes/s**).
 - **Amazon WorkEvents Data Platform**: built the **1-petabyte** warehouse/lake (**AWS Glue, Spark**) for financial reporting & pay computation.
- 🔗 TechStack: **Kotlin, Java, Python, PyTorch, XGBoost, Spark, Kafka, DynamoDB, Redshift, AWS**

SDE, Razorpay, Bengaluru, India

Aug 2020 – Feb 2021

- Built microservices for **P2P Unified Payments Gateway** (Go, Protobuf): **3M+ transactions/hour** at peak, **99.99% uptime, 150ms P99**; customers: **Zomato, Swiggy, Zerodha, Groww** (AWS, GCP).

Software Engineer, Mindfire, Bhubaneswar, India

Aug 2018 – Feb 2020

- Java/Spring full-stack tools; AR product-search microservice (**10K+ daily queries**).

🔧 SKILLS

🔗 **Languages**: Rust, C++, Kotlin, Java, Go, C, Python, TypeScript, JavaScript, Shell Script

🔗 **Systems Programming & Performance**: Lock-Free Data Structures, Concurrency, Memory Safety, Memory Profiling, Cache Optimization, Performance Engineering, Low-Latency Systems, OS Internals, Computer Architecture

🔗 **Networking & Cloud Security**: TCP/IP, UDP, TLS, HTTP/2/3, QUIC, RDMA, InfiniBand, DPDK, eBPF, XDP, Cilium, VPC, NAT, VPN, Load Balancing

🔗 **Distributed Systems & Streaming**: Consensus, Replication, Sharding, Kafka, Flink, Spark, Spark Streaming, Kinesis, CDC, Microservices; gRPC, REST, Protobuf

🔗 **Cloud & Infra**: AWS (EC2, ECS, Lambda, Kinesis, EMR, SageMaker), Azure (AKS), Google Cloud, Kubernetes (EKS, AKS), Docker, Terraform, Kubebuilder, Istio

🔗 **ML Inference & AI**: Deep Learning, Computer Vision, PyTorch, TensorFlow, XGBoost, BERT, LLMs, ONNX Runtime, YOLOv8/v11, CNN, NLP, Edge Inference, MLOps

🔗 **GPU & Parallel Computing**: CUDA, CUDA-Oxide (Rust-to-CUDA), SIMT, GPU Kernel Programming, HPC

🔗 **Observability**: Prometheus, Grafana, Azure Monitor, AWS CloudWatch, Trace & Sampling Data

🔗 **Data & Storage**: DynamoDB, PostgreSQL, Redis, MongoDB, MySQL, Amazon Redshift, Amazon S3

🔗 **Core CS**: Data Structures & Algorithms, System Design, Operating Systems, Linux

🎓 EDUCATION

B.Tech, Computer Science & Engineering, RCC Institute of Information Technology, Kolkata | 2018